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AI GOVERNANCE & INFRASTRUCTURE | CONTROL-LAYER RESEARCH | ECONOMIC MEASUREMENT

Applied-research professional working at the intersection of technical infrastructure, economic measurement, and policy governance. Eight-paper research program under the **Tokenization Systems** brand synthesizes financial economics, econometrics, regulatory interpretation, and governance design around a unifying thesis: regulate and audit the operator and router, not the token. Four papers live on SSRN; four awaiting SSRN approval; one also under peer review at *Frontiers in Blockchain* (Blockchain Economics specialty); causal difference-in-differences study on the SVB stablecoin shock in progress; work submitted to the Fifth Conference on the International Roles of the U.S. Dollar (Federal Reserve Board / Federal Reserve Bank of New York, June 2026). Eight comment letters filed across six federal dockets (April 2026) on the GENIUS Act implementation and AML reform cycles. Claude-native research and knowledge-gathering workflow anchored by ten custom-built MCP servers, a 69-item skill library, and a 15-file living-knowledge architecture.

EXPERIENCE

Senior Investment Analyst, Digital Assets | Borderless Capital

Dec 2019 – Present

- As the firm's first research hire, built the digital asset research function from scratch; lead a four-analyst team producing ~25 long-form publications annually; support diligence and monitoring across ~250 investments (~\$500M AUM, ~10 funds) after sourcing and evaluating 1,000+ opportunities.
- Designed the **Control Layer Intensity Index (CLII)**, a governance-concentration measure across stablecoin gateways (custody segregation, reconciliation integrity, concentration risk), and the **Minimum Viable Equivalence Pack (MVEP)**, a nine-category institutional diligence framework for tokenized products. Both frameworks published on SSRN and adopted in firm diligence.
- Ran Johansen cointegration linking stablecoin supply and Federal Reserve balance-sheet variables across monetary-policy regimes; traced SVB cascading contagion and BUSD routing migration to identify where freeze, routing, and deplatforming authority concentrates; documented gateway concentration rising as volume doubled while unique counterparties fell 25% (2023–2025).
- Synthesize across financial economics, econometrics, regulatory interpretation, and governance design for institutional-investor and policy audiences. Translate tokenized-settlement and gateway-design questions into supervisory, financial-stability, and monetary-policy terms for senior stakeholders and committee use.
- Filed eight comment letters across six federal dockets on the GENIUS Act implementation cycle and the April 2026 AML reform package, applying the control-layer thesis across Treasury, OCC, FDIC, NCUA, and FinCEN. AG19 Recommendation 7 introduces MVEP-gated conditional pass-through, a structural-discipline mechanism that rewards operators whose architectures supply the functional equivalents of deposit protection. Follow-up engagement with Stephen Miran (Federal Reserve Board) and the Ubyx CEO on clearinghouse architecture.
- Forecasted \$1B+ on-chain protocol collapse ~12 months early by mapping the causal chain and defining leading-indicator thresholds. Prevented hostile governance capture on a top-50 Layer 1 via voting-path simulation. Identified and corrected circulating-supply misreporting that had propagated to CoinMarketCap and CoinGecko.
- Convene and instruct across the tokenization research ecosystem: 9 accelerator cohorts (~75 teams taught on mechanism design, regulatory architecture, and protocol measurement); ETHDenver pitch judge; ongoing advisory for early-stage tokenization teams.
- Engineered Claude-native research infrastructure that makes the research and policy throughput above possible: ten custom-built MCP servers spanning on-chain, macro, and developer APIs (FRED, DefiLlama, Nansen Pro, CoinGecko, Helius, Artemis, Spacescope, rated.network, Beaconcha.in, GitHub), a 69-item skill library encoding repeatable research workflows, and a 15-file living-knowledge architecture coordinating Claude.ai, Claude Code, Cursor, and Cowork. Active Claude user since Opus 3.

SELECTED RESEARCH

- **Routing the Dollar.** *Submitted, Federal Reserve Board / FRBNY Fifth Conference on the International Roles of the U.S. Dollar, June 2026. SSRN 6267698.* Johansen estimation documents regime-dependent cointegration between stablecoin supply and Federal Reserve balance-sheet variables ($r = +0.93$ during QE, $r \approx 0$ during tightening, $r = -0.98$ during easing). Evidence is consistent with gateway concentration shaping monetary-policy transmission and operator accountability under stress; three Tier 1 gateways with near-identical Control Layer Intensity scores produced opposite SVB outcomes, and gateway volume doubled while unique counterparties fell 25% (2023–2025).
- **Tokenomics as Institutional Design.** *Under peer review, Frontiers in Blockchain (Blockchain Economics specialty).* A 40-protocol cross-section spanning DePIN, DeFi, infrastructure, and social token categories (tokenized coordination networks where humans and machines collaborate through programmable incentives) shows no significant association between initial token allocation design and governance concentration (Pearson $r = 0.19$, $p = 0.25$, $N = 37$). Sector contrast survives multivariate adjustment for protocol age, valuation, and insider allocation (Mann-Whitney $p = 0.031$, Cohen's $d = 0.96$), and protocols with more insider retention exhibit higher concentration among non-insider holders (Spearman $\rho = 0.54$, $p = 0.001$).
- **Minimum Viable Equivalence Packs.** *SSRN 6363138.* Nine-category institutional diligence standard for tokenized products, anchored on the SVB weekend's cascading infrastructure failures.
- **Adaptive Tokenomics.** *SSRN 6364158.* Systems-engineering stress test with 240-run Monte Carlo ensemble across four adversarial scenarios; adaptive emissions as tail-risk insurance; slashing dominates supply during bootstrapping.
- **Dollar v3: The Control Layer War.** *Published.* How CLARITY and GENIUS provisions attach regulatory obligations to control-layer infrastructure operators rather than to token-level objects.

POLICY ENGAGEMENT

- **NPRM comment letters (April 10–20, 2026), eight filings across six dockets:** Treasury state-similarity (TREAS-DO-2026-0232); OCC original and supplemental (OCC-2025-0372); FinCEN general AML/CFT (FINCEN-2026-0034) and PPSI AML/CFT (FINCEN-2026-0100); banking-agency joint rule (91 FR 18304) filed to OCC, FDIC, and NCUA dockets; FDIC RIN 3064-AG20 (application rule) and RIN 3064-AG19 (substantive rule, with MVEP-gated conditional pass-through in Recommendation 7).
- **Foresight Institute:** AI Ethics and Governance working group (2025).
- **Outreach:** Stephen Miran (Federal Reserve Board); Ubyx CEO (clearinghouse architecture); Uniswap Foundation governance working group (2024).

TECHNICAL

- **Methods:** Johansen cointegration / vector error correction, difference-in-differences (fixest, did, synthdid), event studies, Monte Carlo simulation, concentration analysis (HHI, Gini).
- **Languages:** Python (pandas, statsmodels, scipy, matplotlib), R (fixest, did, synthdid, modelsummary), SQL (DuneSQL, Trino), JavaScript / TypeScript.
- **Data & infrastructure:** Dune Analytics, Nansen Pro, Artemis, DefiLlama, FRED, CoinGecko, Helius, Spacescope, Blockscout, rated.network, Beaconcha.in. Ten custom-built Python MCP servers.
- **Claude-native research and knowledge-gathering:** Claude.ai (strategy, synthesis, editorial), Claude Code (data pipelines, docx XML surgery, econometric estimation), Cursor (site development), Cowork (browser automation); 69-item skill library; 15-file living-knowledge architecture (program state, decision logs, data registries, key findings, known unknowns, framework specs, error-correction logs, cross-reference maps), searched as project knowledge before any new analysis runs.

EDUCATION

High Point University | B.S.B.A., Business Administration (Magna Cum Laude)
Investment Club President | Lingnan University exchange (Hong Kong)

2018